

JAVA METHOD OVERRIDING PROGRAM

Object-Oriented Programming Practical Record

WEEK 7

Question - Method Overriding - Implement RBI as a base class and SBI, ICICI and PNB as derived classes. Demonstrate dynamic polymorphism by displaying the interest rate for the selected bank.

Concept: Inheritance, method overriding and dynamic polymorphism.

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1. Steps for Execution

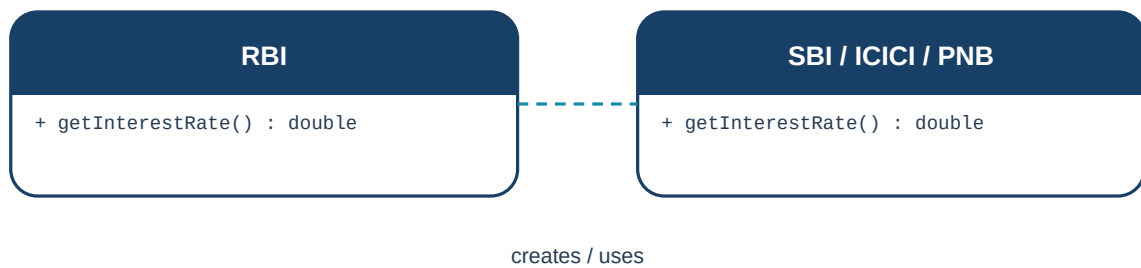
1. Open a Java IDE or command prompt with the Java Development Kit installed.
2. Create a source file named `Week7BankMethodOverriding.java`.
3. Copy the program into the file and save it.
4. Compile it using `javac Week7BankMethodOverriding.java`.
5. Run it using `java Week7BankMethodOverriding`.
6. Provide the required input when prompted and observe the result.

2. Algorithm

1. Create the RBI base class with `getInterestRate()`.
2. Override `getInterestRate()` in SBI, ICICI and PNB classes.
3. Read a bank name from the user.
4. Store the selected child object in an RBI reference and call `getInterestRate()`.

3. UML Class Diagram

The diagram shows the classes and their principal attributes and methods used in this program.



4. Java Source Code

Filename: Week7BankMethodOverriding.java

```
import java.util.Scanner;

class RBI {
    double getInterestRate() { return 4.0; }
}

class SBI extends RBI {
    @Override double getInterestRate() { return 7.0; }
}

class ICICI extends RBI {
    @Override double getInterestRate() { return 6.5; }
}

class PNB extends RBI {
    @Override double getInterestRate() { return 6.0; }
}

public class Week7BankMethodOverriding {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter bank name (RBI/SBI/ICICI/PNB): ");
        String bankName = scanner.nextLine().trim().toUpperCase();
        RBI bank;
        switch (bankName) {
            case "SBI": bank = new SBI(); break;
            case "ICICI": bank = new ICICI(); break;
            case "PNB": bank = new PNB(); break;
            default: bank = new RBI(); bankName = "RBI";
        }
        System.out.printf("%s rate of interest is: %.1f%%\n", bankName, bank.getInterestRate());
        scanner.close();
    }
}
```

5. Representative Sample Output

```
Enter bank name (RBI/SBI/ICICI/PNB): SBI  
SBI rate of interest is: 7.0%
```

Result

Thus, the Java program was written and executed successfully.